CHALMERS

CANDIDATE

DAT232-0001-TOG_2026-08-26

TEST

DAT232 Summer 2026 Omtentamen 2 - L

Subject code	--
Evaluation type	3
Test opening time	26.08.2026 14:00
End time	26.08.2026 18:00
Grade deadline	--
PDF created	23.10.2026 11:40
Created by	Emil Skommargård

CHALMERS | GÖTEBORGS UNIVERSITET

DAT230/DAT232/DIT285 Advanced Requirements Engineering

Re-Exam

Wednesday, August 26

Examiner
Eric Knauss +46 31 772 10 80
Contact person during exam
Nayat Astaiza Soriano +46 70 464 89 43

Allowed tools / material
- English dictionary

General information
The exam consists of three parts:
- Part 1: 30 multiple choice questions, each with one correct answer and worth 1 Point.
- Part 2: One interactive question which is worth 10 Points
- Part 3: Four essay questions, each worth 10 Points. Maximal points can be given if:

  - The answer is correct.
  - The presentation of the answer is readable and clear.
  - The answer is given in English.

Grading
The grades on this exam are based on your total score on the questions:
0 – 39 points: Fail
40 – 55 points: 3
56 – 67 points: 4
68 – 80 points: 5

Review
Will be announced on Canvas.

Part 1: Multiple Choice Questions

This part consists of multiple-choice problems. These problems consist of pairs of Propositions and reasons. For each problem, you shall answer by using one of the following answers:

A Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
B Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
C The proposition is a true statement, but the reason is false.
D The proposition is false, but the reason is a true statement.
E Both the proposition and the reason are false.

Correctly answered problems give 1 point, while incorrect or missing answers give 0 points, regardless if you are partially correct in your answer!

1

Proposition	Reason
According to IEEE Definition of *Requirement* from 1990, a requirement must be explicitly written down, otherwise it is not a requirement.	According to the IEEE Definition from 1990, a Requirement is: (1) A condition or capability needed by a user to solve a problem or achieve an objective. (2) A condition or capability that must be met or possessed by a system or system component to satisfy a contract, standard, specification, or other formally imposed document. (3) A documented representation of a condition or capability as in (1) or (2).

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✓
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

2

Proposition	Reason
Component names, database fields,and similar details in a requirement indicate that it is problem-based.	Considerations of solutions and SW design should be avoided when specifying requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

3

Proposition	Reason
It is crucial to satisfy some issues from each stakeholder group when processing the raised issues from focus groups.	All stakeholder groups have to feel that they get something.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the propositio correct. ✔
- B) Both the proposition and the reason are correct statements, but the reason does n explain the proposition. ✖
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

4

Proposition	Reason
If a stakeholder uses unexpected or ambiguous wording during an interview, this should be ignored during the interview.	Missing information from an interview can easily be provided during analysis.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

5

Proposition	Reason
Actors that are not in the immediate surrounding of the product (the "inner domain") should be excluded from the task descriptions.	Business process re-engineering takes the present user tasks for granted.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

6

Proposition	Reason
An analyst shall document how each business goal is reflected in the requirements.	It is the supplier’s responsibility to reach the customer’s business goals.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

7

Proposition	Reason
For a given domain d: when a customer is inexperienced in d and their supplier is experienced in d, the main goal of requirements engineering should be to reduce cost.	Since the supplier is more experienced, elicitation of customer requirements does not provide value and the potential to satisfy the customer can be maximised if the supplier is just providing their solution directly.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

8

Proposition	Reason
Identification of stakeholders is an important part of requirements elicitation.	Stakeholder needs must be analysed and refined. Without that, they are considered to be "raw requirements", not necessarily being requirements according to definition.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✔
- C) The proposition is a true statement, but the reason is false. ✖
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

9

Proposition	Reason
Symmetry of ignorance is a challenge of requirements elicitation that makes it difficult to map a problem to a solution.	Symmetry of ignorance refers to the fact that requirements engineers should ignore solutions and focus on problems instead.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

10

Proposition	Reason
According to Glinz, all non-functional requirements are quality requirements.	According to Glinz, a non-functional requirement is either an attribute or a constraint of a system.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

11

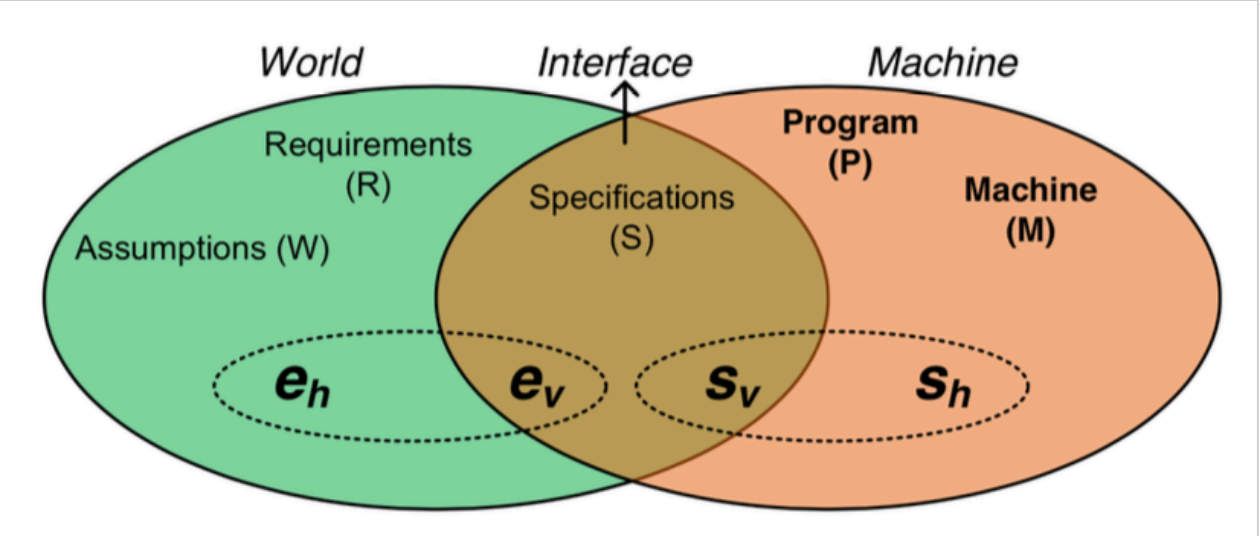
Proposition	Reason
In practice, it is often difficult to distinguish functional and non-functional requirements.	Quality requirements often manifest as extra functionality.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the propositi ✔ s correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✖
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

12



Proposition	Reason
The WRSP model does support architectural decomposition.	The WRSP model defines requirements and specification based on events in the "world" and states in the "machine". These states can be used to describe the logical architecture, thus allowing to map requirements to logical components.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

13

Proposition	Reason
Context diagrams are useful for defining the project scope	Context diagrams show the system under consideration in relation to other external entities such as other systems, data stores, or stakeholders. By showing critical interfaces it can define the context for use cases and task descriptions.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

14

Proposition	Reason
A data model is insensitive to the level of abstraction we work on.	Elements in an E/R model created during requirements analysis will also be relevant in the final product and vice versa.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

15

Proposition	Reason
The event list & function list approach allows to trace between product-level requirements and design-level requirements.	The event list & function list approach establishes many-to-many relationships between domain events and product events.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

16

Proposition	Reason
A feature based approach to documenting requirements facilitates an unambiguous requirements specification.	In a feature based approach, each requirements is one statement about a feature of a system.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✖
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

17

Proposition	Reason
When using screens and prototypes, it is enough to review the screens to validate task support and high usability.	Experience has shown that screens as requirements leads to a smoother programming phase and a higher user acceptance of the product.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✖
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

18

Proposition	Reason
Convergent thinking (to evaluate and choose the best ideas) should be avoided during Creative Problem Solving (CPS).	A core principle of CPS is to defer judgment.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

19

Proposition	Reason
Creativity techniques in requirements engineering should replace elicitation.	Stakeholders cannot provide novel requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✔

Correct. 1 of 1 marks.

20

Proposition	Reason
Functionality is not a quality characteristic.	Quality characteristics include specific quality attributes and performance, but no functional requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✖
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement. ✔
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

21

Proposition	Reason
A taxonomy of system and software quality characteristics helps to identify relevant quality attributes.	Many established quality taxonomies such as ISO25010 and ISO25019 are incomplete.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✔
- C) The proposition is a true statement, but the reason is false. ✖
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

22

Proposition	Reason
Open metric allows to rely on the suppliers expertise with respect to knowing how to measure the quality of the system.	Open metric delegates responsibility for reaching the project goals to the supplier, who is now responsible for the acceptance test of quality requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✖
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

23

Proposition	Reason
Planguage is a template that facilitates testable specification of quality requirements.	While the other fields in the Planguage template provide valuable meta data, it is the Meter field that ensures testability of requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✓
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

24

Proposition	Reason
In prioritization, the ordinal scale should be favored, since it has the highest expressive power of all available scales.	The ordinal scale allows to express that requirement A is 2x as important than requirement B.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

25

Proposition	Reason
Priority grouping is a simple, but often flawed approach to prioritization.	Priority grouping does not require challenging requirements against each other and may lead to too many high priority requirements.

Select one alternative:

A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the propositi ✔ s correct.

B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✖

C) The proposition is a true statement, but the reason is false.

D) The proposition is false, but the reason is a true statement.

E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

26

Proposition	Reason
Even outdated tracelinks are better than no tracelinks.	Outdated tracelinks create additional work, as irrelevant artifacts must be considered when relying on traceability. At the same time, relevant artifacts may be overlooked, since a tracelink has not yet been set. In particular the latter can be critical in many use cases.

Select one alternative:

A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.

B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.

C) The proposition is a true statement, but the reason is false.

D) The proposition is false, but the reason is a true statement. ✔

E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

27

Proposition	Reason
Horizontal traceability refers to tracing across different phases of the waterfall model.	Ramesh and Edwards define horizontal traceability as tracing between artifacts of different types (e.g. tests and requirements).

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✔

Correct. 1 of 1 marks.

28

Proposition	Reason
Pre-requirements traceability supports validation.	Pre-requirements traceability connects requirements to their source or origin, thus establishing whether a requirement is needed.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the propositi ✔ s correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

29

Proposition	Reason
GDT is a specific traceability matrix.	Traceability matrices are often shown with the same artifacts in both dimensions and an indication in a "cell" if the artifacts of this cell's column and row are connected.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✓
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

30

Proposition	Reason
A graphical model that connects business goal 1 with task description 7 is not a TIM.	A TIM is the metamodel that defines which tracelinks between artifacts are allowed and which tracelinks between artifacts are required.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct. ✓
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

31

Proposition	Reason
In change impact analysis, traceability can be used to generate the Candidate Impact Set.	If important tracelinks are missing, the candidate impact set my not include all artifacts of the actual impact set.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✔
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✖

Wrong. 0 of 1 marks.

32

Proposition	Reason
Inspections can provide both validation and verification of requirements.	Inspections are a systematic method for manually reading through specifications.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✔
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

33

Proposition	Reason
Usability tests is a good approach for requirements verification.	During a usability check, the facilitator will guide the user through checking the consistency of requirements.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✓

Correct. 1 of 1 marks.

34

Proposition	Reason
A CRUD matrix supports verification of requirements.	Verification can be expressed as the question "Do the requirements fulfil the stakeholder needs/goals" and a CRUD matrix allows to trace from data requirements to goals.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.
- C) The proposition is a true statement, but the reason is false. ✓
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false. ✗

Wrong. 0 of 1 marks.

35

Proposition	Reason
In large-scale agile approaches, non-functional requirements should not be managed on team level.	Non-functional requirements are usually cross-cutting, thus, they affect multiple components, functional requirements, and teams.

Select one alternative:

A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the propositi ✔ s correct.

B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✖

C) The proposition is a true statement, but the reason is false.

D) The proposition is false, but the reason is a true statement.

E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

36

Proposition	Reason
A user story is more than just the text, it should also include confirmation, i.e. describing its acceptance test.	The most common template for a user story (As a <role> I want <feature> so that <value>) ensures that a user story is testable.

Select one alternative:

A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.

B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.

C) The proposition is a true statement, but the reason is false. ✔

D) The proposition is false, but the reason is a true statement.

E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

37

Proposition	Reason
Agile methods promise to mitigate communication gaps in requirements engineering.	Agile RE promotes regular interaction with customer and between teams.

Select one alternative:

A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the propositi ✔ s correct.

B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✖

C) The proposition is a true statement, but the reason is false.

D) The proposition is false, but the reason is a true statement.

E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

38

Proposition	Reason
During requirements analysis, anonymisation is no longer a concern.	Anonymisation (defined as a component of confidentiality involving not revealing any data that can be used to identify subjects) should happen during elictation.

Select one alternative:

A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.

B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition.

C) The proposition is a true statement, but the reason is false.

D) The proposition is false, but the reason is a true statement. ✔

E) Both the proposition and the reason are false.

Correct. 1 of 1 marks.

39

Proposition	Reason
Ambiguous requirements can cause ethical problems.	Ambiguity in requirements often leads to ambiguous or biased interpretation, however, biases and fairness are important ethical consideration.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the propositi ✔ s correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✖
- C) The proposition is a true statement, but the reason is false.
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

40

Proposition	Reason
Many modern vehicles (not only those with self-driving capabilities) take pictures and videos of their surroundings, especially of humans that approach them while they are parked. This can be considered as an ethical problem.	Imagine a school child approaching a parked vehicle. The vehicle taking pictures and sharing them with an unknown group of recipients can be considered a violation of the EU ethical guideline of Nonmaleficience.

Select one alternative:

- A) Both the proposition and the reason are correct statements. In addition, the reason explains the proposition in a correct way, i.e. the reason explains why the proposition is correct.
- B) Both the proposition and the reason are correct statements, but the reason does not explain the proposition. ✖
- C) The proposition is a true statement, but the reason is false. ✔
- D) The proposition is false, but the reason is a true statement.
- E) Both the proposition and the reason are false.

Wrong. 0 of 1 marks.

Part 2: Interactive questions

41 This part of the exam consists of a match-assignment where different alternatives shall be matched with the most suitable, proposed, alternative. A correct answer gives 1 points, while incorrect or missing answers gives 0 points even if you are partially correct in your answer!

Match the ten problems described below with the most suitable alternative problem source. The same problem source may be the answer in more than one row of the table; some problem sources may not be the answer in any row. Only one problem source shall be selected on each row!

Please match the ten problems described below with the most suitable alternative problem source.

	Programm Error	Requirements Defect	Usability Problem
1) It is impossible to carry out the user registration task.	○	○ ✓	○ ✗
2) The user erroneously believes that the registration task is completed.	○	○	○ ✓
3) The user can register after lengthy attempts.	○ ✗	○	○ ✓
4) The user registration functionality does not work as the programmer had intended.	○ ✓	○	○
5) The user cannot figure out which discount code to give the customer, although he knows which field to use.	○	○	○ ✓
6) The system does not provide the necessary fields to store user data.	○	○ ✓	○
7) The system crashes when the user attempts to store user data.	○ ✓	○	○
8) The system does not offer an option to remove an item from the shopping cart.	○	○ ✓	○
9) The user is asked to include between 2 to 4 digits in the new password. The program only checks whether the password includes any digits.	○ ✓	○	○
10) The user thinks he has updated the password successfully, but the system requires the user to confirm the new password to complete the updating task.	○	○	○ ✓

Partially Correct. 8 of 10 marks.

42

Please match the requirements in the table below with the most suitable alternative level.

	Goal-level	Domain-level	Product-level	Design-level
The product shall support estimations that match actual scores within a standard deviation of 1%.	✓			
For all fields, the user shall be able to select the value from a drop-down list.				✓
The product screens shall look as shown in appendix xx.				✓
The menu points and buttons shall work as described in appendix yy.		✗		✓
The product shall handle the booking task.	✗	✓		
The system shall allow efficient usage with one hand only.	✗		✓	
The system shall have a mini keyboard with start/stop button.				✓
The product shall have a function for storing new items based on their associated keywords.			✓	
The product shall have a function for retrieving items based on specified keywords.			✓	
The product shall support the user registration task.		✓		

Partially Correct. 7 of 10 marks.

Part 3: Essay questions

For each topic below, write an essay that includes all of the listed concepts within the given maximum number of pages. The essays are graded based on

1. how well the topic is described using the concepts in the list under each topic (thus judging the general flow and argumentation), and
2. how well the concepts in the list are described and/or exemplified in the essay (thus, points are given for each concept that is mentioned and explained in sufficient depth).

43

For the topic below, write an essay that includes all of the listed concepts within the given maximum number of words. The essays are graded based on

1. how well the topic is described using the concepts in the list under each topic (thus judging the general flow and argumentation), and
2. how well the concepts in the list are described and/or exemplified in the essay (thus, points are given for each concept that is mentioned and explained in sufficient depth).

Topic: Completeness of Elicitation

Concepts:

- One example of an early-phase elicitation technique
- One example of a mid-phase elicitation technique
- One example of a late-phase elicitation technique
- tacit knowledge
- effect of asking "why"

Write your answer in the box below. Changes are saved automatically.

Elicitation:Collecting the requirements from various stakeholder .Some requirements are hidden or missed .By doing elicitation we can find missed requirements.

Early phase :**Interviews or group Interview---Both are different.**

Here we interview specific people about the system.we also interview different stakeholders what they are expecting to be in the appliction.Example for online grocery application .we interview like some users who order groceries online will ask specific questions like per week how many times they order.Delivery boys.

In group interviews some people will be dominating.

Mid Phase:Questionnaires

We create some questions and send some of the participants and ask them to fill the questions.If we

Questionnaires should be designed by the people who have complete knowledge and They know problems of the system.If they don't have knowledge on the system they should not prepare questions because while filling by participants they will not know whats right and wrong.

Late Phase :Prototyping

Prototypes are usefully while validation time

Prototypes can be digitally or on paper.

where we design mock application.By this analysts and developer no which requirements are missed.

While testing prototype are usefull.

Tacit knowledge: Stakeholder cannot describe what they want to be in the system because by using application they can use without anybody help but application they cannot explain what they are doing.Because developer know technical terms customer know business strategy like they don't have common language .So they cannot explain what they are expecting in the system.My elicitation techniques like observations,interview we can get information from them.

Asking why: Developer ask why questions for example in this scenrios

If customer ask he want reviews for the groceries
while this time developer negotiate with the customer . why that is needed now.If it is important and if that need they will add later

Words: 291

Answered.

44 For the topic below, write an essay that includes all of the listed concepts within the given maximum number of words. The essays are graded based on

1. how well the topic is described using the concepts in the list under each topic (thus judging the general flow and argumentation), and
2. how well the concepts in the list are described and/or exemplified in the essay (thus, points are given for each concept that is mentioned and explained in sufficient depth).

Topic: Understanding Quality Requirements

Concepts:

- Quality Grid
- Planguage
- Open Metric
- Constraint
- Attribute

Write your answer in the box below. Changes are saved automatically.

Quality Grid: Quality grid is specified quality for every functionality like security,usability ,reliability etc

So developers know for each functionality which quality is important and most foused on the specfic quality

Operation:During the system use

Usability,security

Revision :Malignance of the system

Maintainability

Transpiration:quality of when login in other platforms like Laptop

Portability

Ex:For banking application

for payment functionality security is very important.

P language :Planned language

P language is a systematic and specific way test and measure the quality requirements .In P language we us keywords like scale,current value,maximum expected,goal ,wish.

In p language the vague requirements are written clearly with testable and measurable value

Open metric: Developer know the requirement is important but don't know the targeted value.The value is left open and described late by supplier or further analyses

Ex:Weather forecast

Constraint: Constraint is a requirement.But like functional it does not specify functionality. Supplier tell them Just they developer know only details but not entire requirement .There will be som may constraints.

Ex:Legal constraint

budget constraint

Th budget should not cross $1000.It should be with in the the budget

Technical constraint

Attribute:Measure the quality requirements with specific value

Ex:usability,reliability,maintainbility

When user searches for specific product .it should display in 2 seconds
It should write concrete value and how it behave .

Words: 212

Answered.